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Space experimental results of STARS-C CubeSat to verify tether deployment in orbit

Yoshiki Yamagiwa[a,*], Tatuya Fujii[a], Kenji Nakashima[a], Hiromu Oshimori[a], Taiki Okino[a], Shogo Komua[a], Shoko Arita[a], Masahiro Nohmi[a], Yoji Ishikawa[b]

[a] Shizuoka University, Hamamatsu, Shizuoka, 432-8561, Japan
[b] Obayashi Co., Minatu-ku, Tokyo, 108-8502, Japan

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ABSTRACT

STARS-C was the first trial of a tether deployment in orbit using a CubeSat to certify and acquire data for the important technology of space elevators. STARS-C is 2U in size and consists of a mother satellite (MS) and a daughter satellite (DS). It is designed to deploy a 100-m aramid fiber tether between the MS and DS. STARS-C was launched on December 9, 2016, released from J-SSOD on December 19, 2016, and operated until its re-entry on March 2, 2018. Unfortunately, the transmission conditions were so poor that detailed data on tether deployment could not be obtained. However, we found that STARS-C rotated from its JAPOS-supported optical data detected from the ground, and this rotation largely influenced the tether dynamics during and after deployment. We also estimated the tether deployment state from the STARS-C altitude history data, the satellite condition from the optical data, and additional ground tests.

1. Introduction

The space tether is a promising space system for many useful applications, such as debris removal, space elevators, and formation flights. As the first step toward the use of space tethers, their smooth deployment in space is indispensable. An understanding of tether dynamics and control during deployment is necessary for smooth deployment. Some experiments on tether deployment in space [1–3] have been attempted in the past 10 years. The tether deployment of STARS was a few tens centimeters for 5 m tether deployment trial. Direct data on tether deployment of STARS-II could not be obtained because of a transmission problem, so such data were estimated on the basis of altitude drop data. In the T-Rex experiment using a sounding rocket, the tether deployment was 133 m out of 300 m and the rebound was happened although the valuable tether dynamics data were obtained by camera and GPS. Thus, a fully successful tether deployment in space has not yet been achieved in Japan. Worldwide, there have been some successful tether deployment experiments in space, such as SEDS, TiPS, and YES [4–6]. Further accumulation of data, however, is needed to develop an accurate model of tether dynamics during tether deployment.

In light of this background, we proposed a tether deployment experiment in space using a CubeSat called STARS-C (Space Tethered Autonomous Robotic Satellite - Cube) [7] that solves problems based on past experiences. With the advantages of a short development term (1–2 years) and low cost, a CubeSat is suitable for performing risky space experiments with original and new systems such as space tethers. STARS-C focused on deploying a tether to allow the study of tether dynamics during deployment, with the goal of developing a smooth tether deployment design. This was the first project officially accepted in Japan to develop the important basic technology of space elevators.

2. Specification and mission sequence of STARS-C

The purpose of STARS-C is to demonstrate the smooth deployment of a 100-m tether in space and to obtain tether dynamics data for the future design of a tether deployment system. STARS-C was selected in September 2014 as a satellite to be released from the “Kibo” ISS module and was delivered to JAXA on November 7, 2016.

2.1. Satellite specification

Fig. 1 illustrates STARS-C. It is 2U in size. It consists of a mother satellite (MS) and a daughter satellite (DS) and is designed to deploy between them a 100-m tether composed of aramid fiber. The diameter of tether is 0.4 mm and the total mass of 100 m tether is 18 g. The MS

* Corresponding author.
E-mail address: yamagiwa.yoshiki@shizuoka.ac.jp (Y. Yamagiwa).

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Nomenclature

a	semi-major axis of satellite orbit
Δa	altitude drop per orbit
A_s	effective cross section of satellite vertical to the flight direction
C_d	drag coefficient of the satellite
C_{ds}	drag coefficient of the satellite body
C_{dt}	drag coefficient of the tether
e	eccentricity of the satellite orbit
f	true anomaly of the satellite orbit
F_C	centrifugal force by satellite rotation
F_G	gravity gradient force
F_R	tether deployment resistance
F_S	spring elastic force by tether coiling
g_0	gravity acceleration
G	gravitational constant
k	spring constant
k_t	total spring constant of series spring
l	natural length of coil shaped tether
l_h	width of STAR-C
l_w	height of STARS-C
L_{tet}	deployed tether length
L	angular momentum
m_d	mass of daughter satellite
m_m	mass of mother satellite
m_s	mass of satellite ($=m_d + m_m$)
m_w	mass of weight
M	Earth mass
r_d	distance from the center of rotation to daughter satellite
r_m	distance from the center of rotation to mother satellite
T	period of satellite orbit
x	distance between mother satellite and daughter satellite
v_d	relative velocity of daughter satellite to mother satellite
v_0	initial velocity of daughter satellite to mother satellite
y	elongation of coil shaped tether
y_m	elongation of coil shaped tether for weight mass m_w
θ	rotation angle of satellite to tether deployment direction
ρ	atmospheric density
ω	angular velocity of satellite rotation
ω_s	angular velocity of the orbit of satellite
ω_0	initial angular velocity of satellite rotation

and DS share common subsystems, including power, communication, and command and data handling systems. They also have a tether unit with a spool and reel mechanisms as a mission system and solar paddles with antennas. Fig. 2 shows a CAD model of the construction of STARS-C and specifications. Fig. 3 shows photographs of the STARS-C FM. As the power source, two 35.4 mm × 56.2 mm × 7.2 mm Li batteries (nominal voltage: 3.7 V, nominal capacity: 1250 mAh), which was charged by eight 40 mm × 80 mm TJ (triple junction) GaAs solar cells (maximum power: 1.3 W, nominal power:1.2 W), were used for each MS and DS.

2.2. Scheduled mission sequence

First, the space experiment performed with STARS-C was scheduled in four steps (Fig. 4) after STARS-C was released from J-SSOD (JEM Small Satellite Orbital Deployer) on Kibo [8,9]. (1) The solar paddles are deployed and the communication link between the satellite and the ground station is verified. (2) The attitude of the satellite is controlled by magnetic torquers to ensure that tether deployment is in as much of a vertical direction to the ground as possible. (3) The MS and DS are separated and the tether is deployed. In this step, the lock wire is cut using a nichrome wire-heating cutter by sending a command from the ground for the satellites to separate using the spring force in the MS. During deployment, the tether tension is adjusted by controlling the reel rotation speed. (4) Finally, the tether dynamics are controlled using the reel mechanism. During these steps, the tether dynamics are observed by cameras and GPSs on the MS and DS.



Fig. 1. Image of STARS-C.

3. Operation of STARS-C

STARS-C was launched on December 9, 2016, and released from J-SSOD on December 19, 2016, (Fig. 5), then operated until reentry on March 2, 2018. The outline of STARS-C's operation was as follows.

December 2016~: Adjustment of ground station (orientation of the antenna, received frequency, transmit power) to find the optimal transmission conditions.

April 2017~: the conditions of signal reception from the satellite were improved.

June 2017~: transmission of the heater cutter command for tether deployment began, with optical observation from the ground.

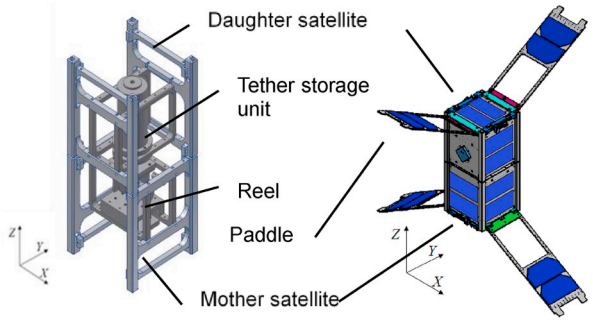
September 2017~: the heater cutter command for tether deployment was transmitted without satisfying the optical conditions from the ground.

Mid-December 2017: the signal was lost.

March 1, 2018: the signal was detected again.

March 2, 2018: reentry (the reentry time estimated by USSTRATCOM was 14:56 UTC, March 2, 2018).

STARS-C survived until its reentry, but the transmission quality during its operation was not good. The ground station received the CW beacon from MS almost constantly, but received that from DS for only a few moments. FM packet transmission from MS was also received sometimes, but it was weak and intermittent, so the data could not be decoded. STARS-C also had difficulty receiving commands from the ground station. By regulating the ground station, the signal reception condition was better than that at the initial situation just after satellite release, although these problems remained until reentry. One possible



Dimensions; 100×100×227[mm] (total of MS and DS),
Tether length; 100m
Mass; MS: 1.4[kg], DS: 1.1[kg], Total: 2.5[kg]
Communication; Amateur radio
(Uplink; 145[MHz] bands, 1200[bps],
Downlink; 437[MHz] bands, 1200/9600[bps]

Fig. 2. Construction and specifications of STARS-C.

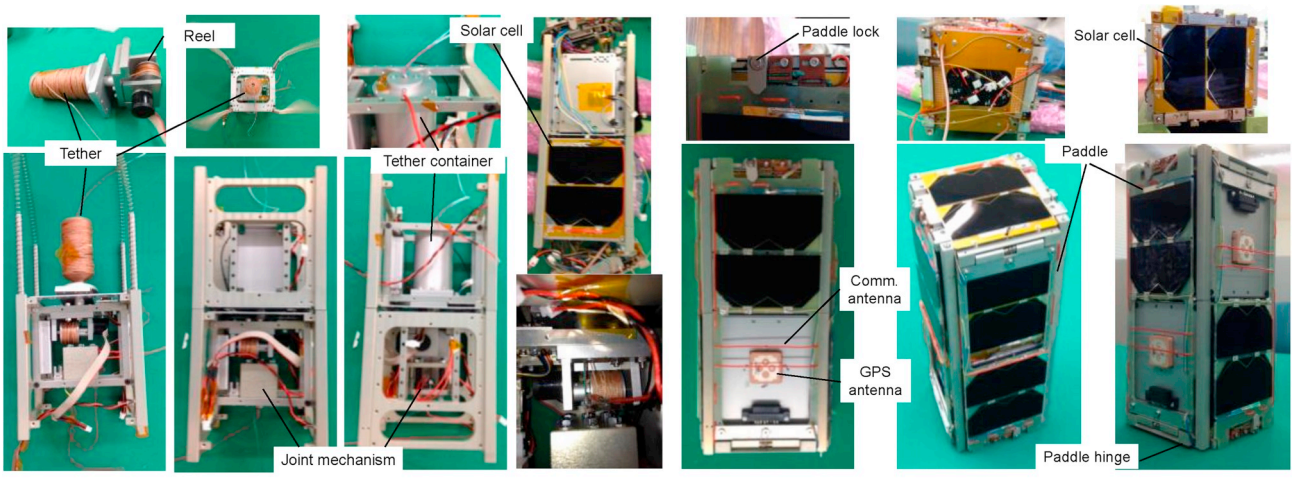


Fig. 3. Photographs of STARS-C FM.

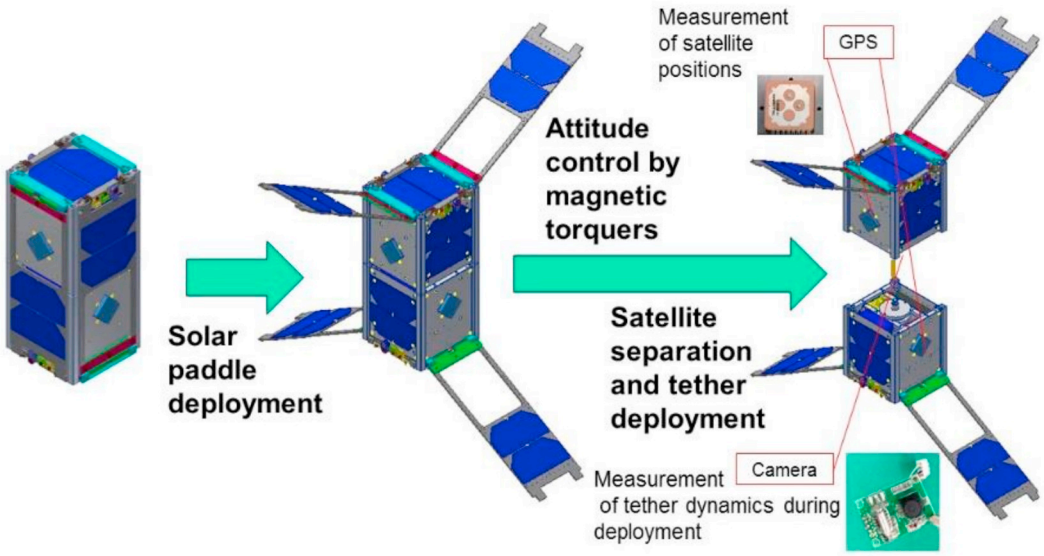


Fig. 4. Mission sequence of STARS-C.

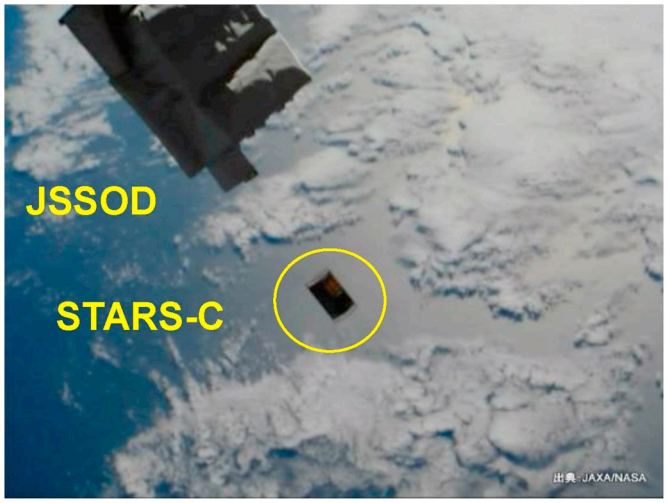


Fig. 5. Release of STARS-C from J-SSOD. (Courtesy of JAXA).

reason for these problems is that the solar paddles were not deployed. In the design, the paddles are deployed 30 min after satellite release from Kibo by the automatic operation of the heater cutter of paddle's fixed wires, and we also verified its operation on the ground. However, it is considered that the fixed wires would not be cut completely in real situations. The paddles were equipped with antennas, which did not work well when the paddles were not deployed. Another possible cause is satellite rotation, as shown later. The direction of the antennas was changed at all times, and this made it difficult to receive commands.

In these conditions, we also tried to detect STARS-C by an astronomical telescope to observe the tether deployment with the cooperation of astronomical observatories in Japan (JAPOS; Japan Public Observatory Society) because it was so difficult to acquire direct data on tether deployment. The results of this optical observation of STARS-C are shown in the next chapter.

4. Results of optical observation from the ground

We tried to observe STARS-C optically by an astronomical telescope to compensate for transmission problems. It was very difficult to detect a CubeSat-sized object by using the telescopes of astronomical observatories. In addition, the daily optical observation window from the ground was brief, as it was limited to the time when sunlight hit the satellite at night under clear skies. However, we succeeded in taking STARS-C pictures 12 times and shooting video footage once. Fig. 6 shows a sample STARS-C picture taken just after the release from Kibo.

We also can estimate the rotation of STARS-C from the video footage. Fig. 7 shows the analytical result of luminance variation of STARS-C from the movie obtained by the Toyama Astronomical Observatory (location; lat. 36°39′21.1″, long. 137°6′5″) on April 12, 2017. The total length of video footage was 110-s. The data shown in Fig. 7 is a part of the footage. The luminance was analyzed by the astronomical image processing software Makali'i [10]. When we split this variation into certain times (designated by the blue, orange, red, yellow, and green lines), we can see that the luminance of STARS-C changes in a certain period. On the other hand, STARS-C luminance is changed by the surface where the sunlight hits, because each surface of STARS-C consists of different materials and has a different optical reflectance, as shown in Fig. 8. So we can find the variation of satellite luminance to analyze which surface of satellite faces a certain direction with respect to a fixed direction during one rotation around a certain satellite axis. When the variation of brightness of STARS-C around each satellite axis is compared with the luminance data observed, the variation shown in Fig. 9 reveals the qualitatively best match (the red line indicates the part of observed data shown in Fig. 9, and the blue line is the possible brightness variation by the analysis when the satellite rotates around the Y axis) (the red line is the observed data, and the blue line is the possible brightness variation when the satellite rotates around the Y axis). We can estimate that STARS-C was rotated with a period of about 4 s around the Y axis of the satellite, and that the rotation was in the pitch direction of the orbit. Of course, another possibility is that STARS-C rotates around an axis other than the X, Y, or Z axis, but a more precise specification is difficult from the obtained data. The cause of this rotation cannot be identified, but it is considered that either an unbalance of force was exerted on the satellite surface at the release from J-SSOD or that another effect, such as solar radiation pressure, caused the rotation.

5. Estimation of tether deployment state

The transmission of the FM packet including the GPS and camera data was of insufficient quality, so we tried to estimate the tether deployment state from the data of STARS-C altitude history as we did in the case of STARS-II. A more accurate estimation is possible than was the case with STARS-II because we have information on the satellite flight condition observed from the ground as shown in the preceding chapter. Because the side cross section of the fully deployed tether (0.4 mm diameter, 100 m length) is 0.04 m^2, which is about twice that the 0.022 m^2 side cross section of the satellite (MS + DS), the orbit descent speed increases significantly when the tether is deployed. Fig. 10 shows the variation history of STARS-C altitude compared with another 2U-sized CubeSat, AOBA-VELOX-3, released from Kibo in the same period. AOBA-VELOX-3 was a 2U-sized CubeSat and the same size (100 mm × 100 mm × 227 mm) as STARS-C. This altitude history of satellites was obtained from the basic relationship between the semi-major axis of the satellite orbit a and the period of its orbit T shown in equation (1) when the circular orbit was assumed, and by using TLE data by NORAD [11] for the period of the satellite orbit.

$$T^2 = \frac{4\pi^2 a^3}{GM} \tag{1}$$

The time history is shown as the time from release, although in fact AOBA-VELOX-3 was released one month after STARS-C. The figure shows that the altitude drop of STARS-C rapidly increased around 250 days after release. This rapid altitude drop was obvious in comparison with AOBA-VELOX-3. The altitude drop can be obtained from equations (2) and (3) when only the atmospheric drag is considered [12].

$$\Delta\{a(1-e)\} = -\left(\frac{A_s}{m_s}\right) c_d \rho a^2 (1-e)^2 \int_0^{2\pi} \frac{(1+e^2+2e\cos f)^{\frac{3}{2}}}{(1+e\cos f)^2}(1-\cos f)df \tag{2}$$



Fig. 6. Observation of STARS-C by astronomical observatory. (courtesy of Kurashiki Science Center).

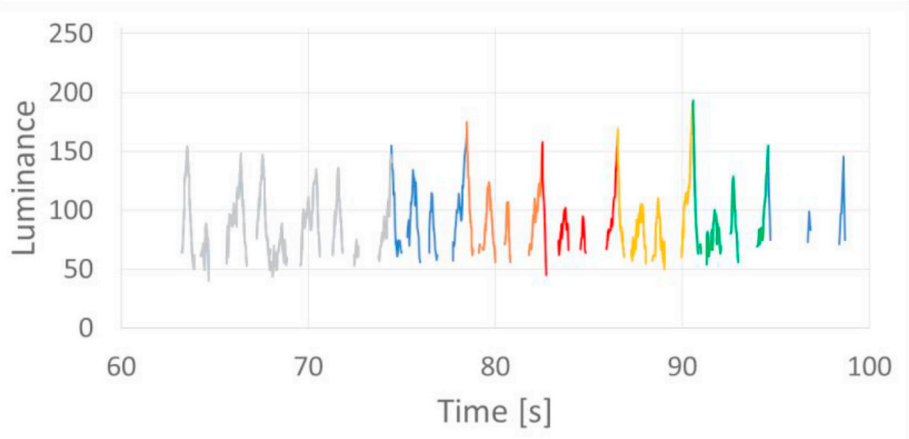


Fig. 7. Variation of luminance of STARS-C analyzed from optical observation from the ground. (from video taken by Toyama Astronomical Observatory).

$$\Delta\{a(1+e)\} = -\left(\frac{A_s}{m_s}\right)c_d\rho a^2(1+e)^2\int_0^{2\pi}\frac{(1+e^2+2e\cos f)^{\frac{3}{2}}}{(1+e\cos f)^2}(1+\cos f)df \quad (3)$$

When the circular orbit is assumed, the altitude drop per orbit, Δa, can be rewritten as

$$\Delta a = -\left(\frac{A_s}{m_s}\right)C_d\rho a^2\cdot 2\pi \quad (4)$$

For STARS-C, m_s is 2.5 kg. The value of a and Δa can be obtained from the altitude history of STARS-C based on TLE data shown above. As shown in nomenclature, A_s is the effective cross section of satellite vertical to the flight direction [m^2], C_d is the drag coefficient of the satellite. We wish to determine the A_s value to find how long the tether was deployed, but we evaluate $C_d \times A_s$ value because the C_d value is different for the satellite body than for the tether. Fig. 11 shows the value of $C_d \times A_s$ calculated considering the variation of atmospheric density with altitude from the altitude history. The data shown in

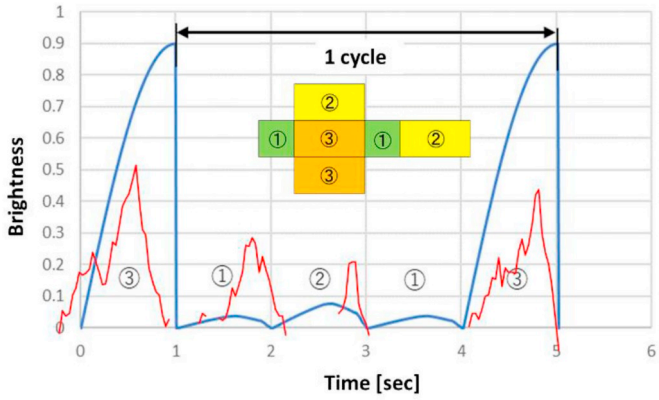


Fig. 9. Comparison of STARS-C brightness.

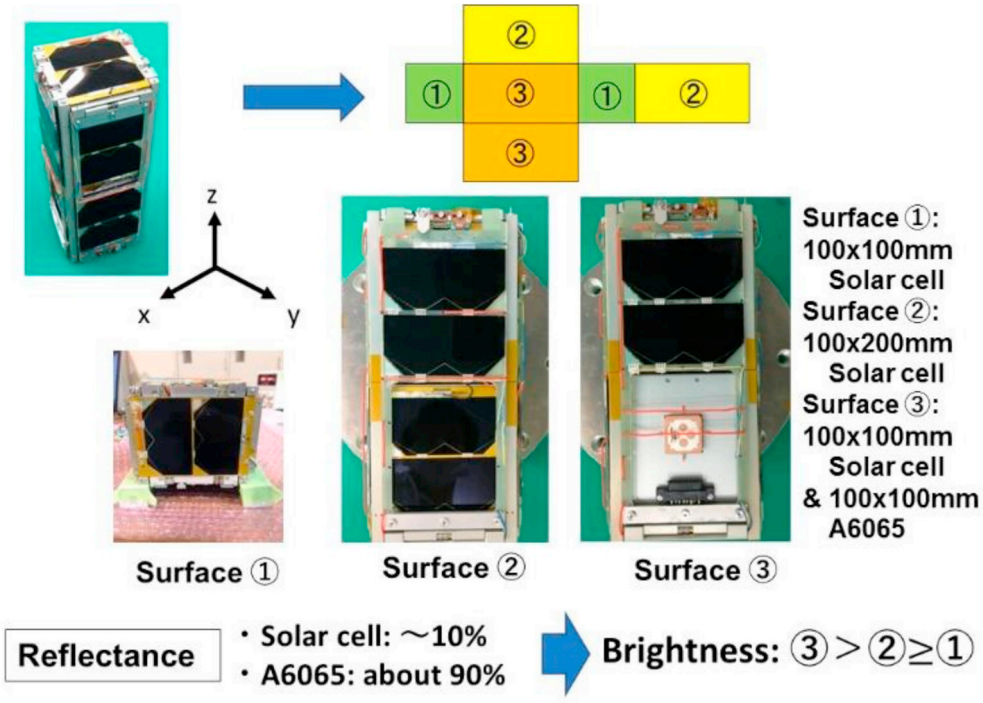


Fig. 8. Optical condition of STARS-C surface.

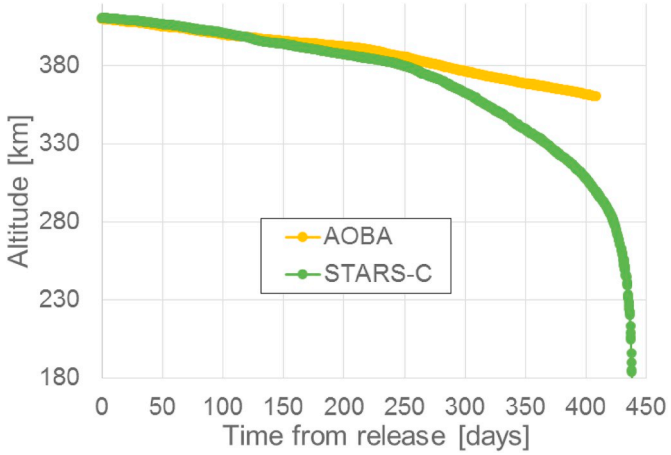


Fig. 10. Variation history of STARS-C altitude.

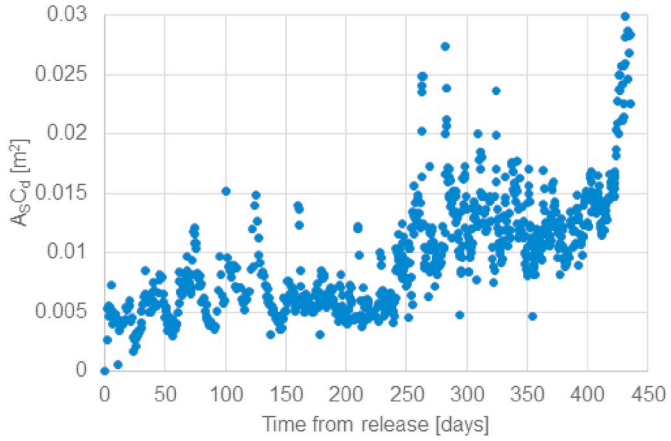


Fig. 11. Variation of drag coefficient C_d × satellite cross section A_s value.

Chronological Scientific Tables [13] which was based on U.S. standard atmosphere 1976 was used as atmospheric density with altitude. We can see the satellite condition obviously changes after about 250 days from release. Table 1 shows the ratio of the average value of $C_d \times A_s$ from before to after 250 days (except for the last phase of more than 420 days). We can see that the $C_d \times A_s$ value after 250 days from release was 1.98 times that from before 250 days.

The shapes of STARS-C at the release and at the solar paddle deployment are shown in Fig. 12. From the side cross section at each shape of the satellite and tether cross section (0.4 mm × length), Table 2 shows the ratio of the $C_d \times A_s$ value when the drag coefficients of the satellite (cube) C_{ds} and tether (cylindrical shape) C_{dt} are assumed as shown in the table based on references [14,15]. The side cross sections are the averaged values considering the rotation of the satellite. STARS-C was designed so that the solar paddles would fully deploy when the MS and DS were separated even if the lock wires of the paddles were not cut. Fig. 13 shows the history of satellite inner time in the operation. The satellite inner time is the time counted from the point when the satellite power is recovered after the power has been down during operation, and its data are included in CW beacon. So the reset of satellite inner time means that the power of the satellite had been downed. This figure shows that there was no reset of satellite inner time after 250 days from release. This means that the power condition improved after 250 days from release. This is proof of the full deployment of the solar paddles after 250 days from release. But the difference in the $C_d \times A_s$ value from before to after around 250 days from release cannot be explained only by the solar paddle deployment. We can see that the difference in the $C_d \times A_s$ value from before to after about 250 days after release is consistent with the condition when the tether was deployed about 30 m in addition to the full deployment of the solar paddles. So it is estimated that STARS-C received the tether deployment command after 250 days from release and that the tether deployment mission was performed, but the deployment length was only about 30 m.

6. Discussion of the tether deployment conditions

As shown in the previous chapter, the tether deployment mission was performed, but the deployment length was only about 30 m. In this chapter we try to suppose how the tether was deployed.

As shown in chapter 4, STARS-C rotated, so we suppose that this rotation significantly influenced tether deployment. In addition, the tether was shaped like a coil in the real state, as shown in Fig. 14, because the tether was wound around the spool, so it is considered that the spring elastic force of the coil shaped tether also influenced tether deployment.

Fig. 15 shows the variation of the tether tension by centrifugal force acting on MS and DS for tether length if the angular momentum of STARS-C rotation was conserved and was transformed to the rotation of the whole system connecting the MS and DS by the tether. The analytical method is shown in Appendix A. The value of the tether tension by centrifugal force decreases rapidly as the distance between the MS and DS increases. The value is higher than that of the gravity gradient force (1×10^{-4} N maximum) during the initial tether deployment until about 5 m, but it is quite small compared to the spring force in the satellite at the satellite separation (27 N). So the deployment was mainly performed by the spring force of the springs in the satellite.

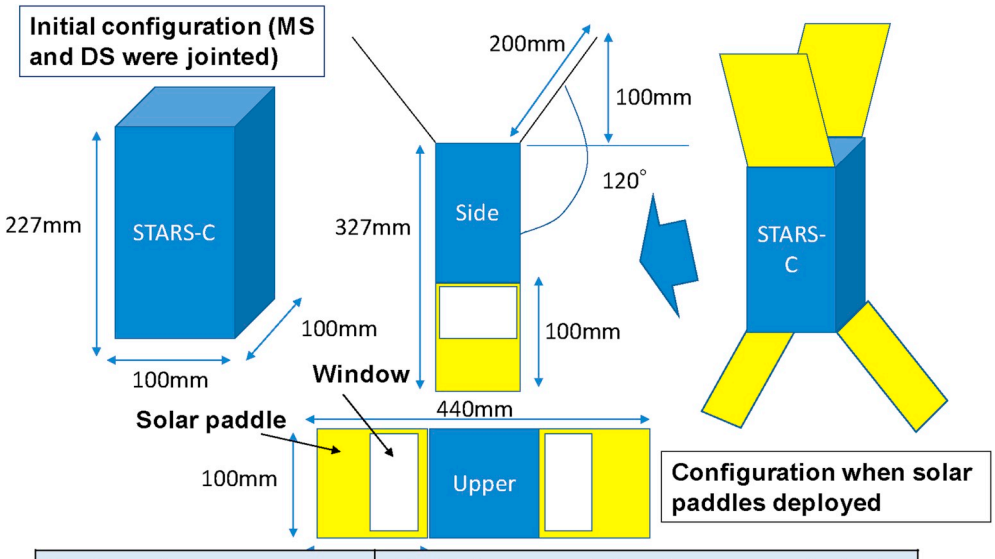
The spring force by tether coiling was measured in an additional experiment on the ground (Appendix C), and Fig. 16 shows how the spring force varies with tether deployment length. The spring force by the coiling is exerted in the last phase of tether deployment when the tether is stretched by the breaking force from the residual deployment speed. The spring force is also more than two orders larger than the gravity gradient force, so it is considered that the spring force contributed to the tether deployment motion in the final phase of the deployment.

Fig. 17 shows the deployment resistance of the tether. The experimental method of tether deployment resistance is shown in Appendix B. The deployment resistance was caused by the contact between each tether and the exit hole of the tether in the tether container. The deployment resistance was more than one order higher than the maximum gravity gradient force. It was lower than the centrifugal force at the satellite separation and one order lower than the spring force by tether coiling. But the centrifugal force and the spring force by tether coiling are exerted only at the initial and final phases of tether deployment, respectively, so the tether deployment motion was mainly dominated by the deployment resistance force. It is also considered that the MS and DS itself each rotated by the conservation of angular momentum of satellite rotation when the tether tension was low, and the tether deployment resistance increased by changing the contact condition of the tether at the container exit and also with the surfaces of satellites. Fig. 18 shows that the variation in tether-deployment resistance with the satellite-rotation angle. The deployment resistance becomes more than one order higher when the satellite rotates more than 180°.

Based on these additional data on tether-deployment resistance, one possible situation of tether deployment in the STARS-C mission was that

Table 1
Average value of $C_d \times A_s$.

Day from release [days]	Ratio of $C_d \times A_s$
0–250	1
250–400	1.98



Configuration	Projection area on the plane perpendicular to the flight direction (cm^2)		
	Max.	Min.	Average
Initial	227.0	100.0	163.5
Solar paddles deployed	270.0	277.0	273.5

Fig. 12. Shape variation of STARS-C.

Table 2
Ratio of $C_d \times A_s$ values for different satellite conditions.

Day from release	Condition	Ratio of $C_d \times A_s$		
		C_{ds} = 2.4, C_{dt} = 2.2	C_{ds} = 2.4, C_{dt} = 2.1	C_{ds} = 2.4, C_{dt} = 2.0
0–250 days	Initial state	1	1	1
250–400 days	Full paddle deployment	1.67	1.67	1.67
	Full paddle deployment +30 m tether deployment	2.01	1.99	1.98

the tether deployment was stopped by the rapid decrease in deployment speed in order to increase the deployment resistance with the rotation of the MS and DS. How long the tether is deployed by these forces is determined by solving the equation of motion of the satellite including the deployment-resistance force, the spring force by tether coiling, and centrifugal force. Fig. 19 shows the variation in satellite separation velocity with satellite distance when the MS and DS each rotate 0–90°, 0–180°, 0–270°, 0–360°, and 0–380°, assuming the average deployment resistance of these angles. The analytical method is shown in Appendix D. The results show that the tether deployment stops at about 30 m if the MS and DS each rotate 0–380°.

On the other hand, another possibility is that the MS and DS rotated less than we estimated, because the rotation of the MS and DS would also be suppressed by the centrifugal force produced by the rotation of

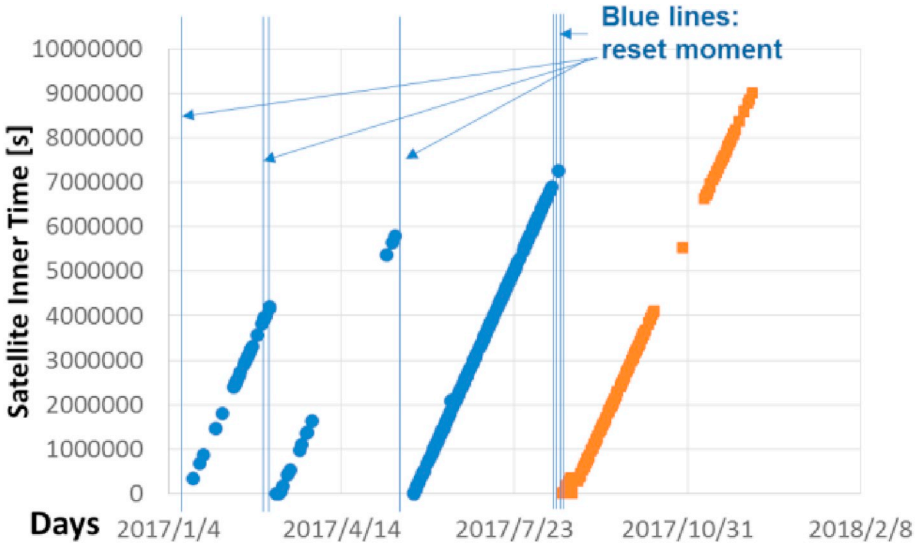


Fig. 13. History of satellite inner-time in the operation.



Fig. 14. Remaining coiled shape of tether.

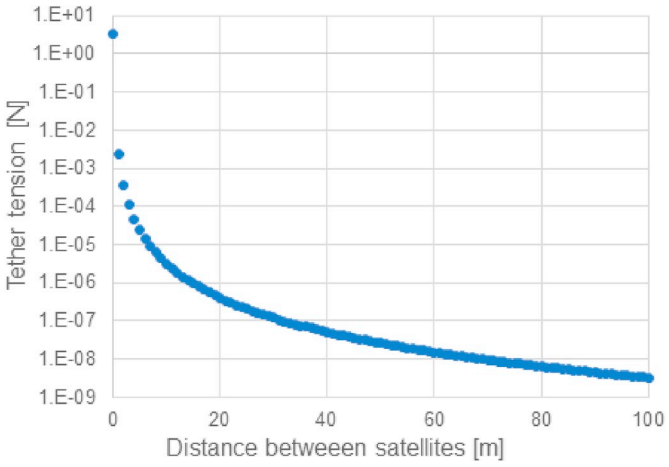


Fig. 15. Tether tension by centrifugal force with STARS-C rotation.

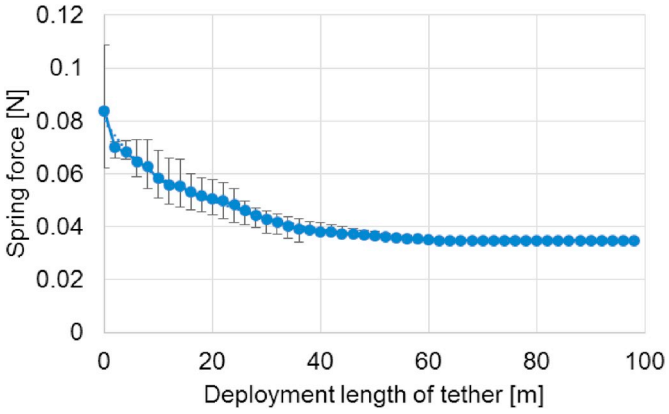


Fig. 16. Spring force by tether coil shape.

the whole system connecting the MS and DS by the tether, and the tether was once fully deployed. In such a case, the following sequence will be considered (Fig. 20). The DS and MS would rebound by the residual tension and by the spring force of tether coiling in the final

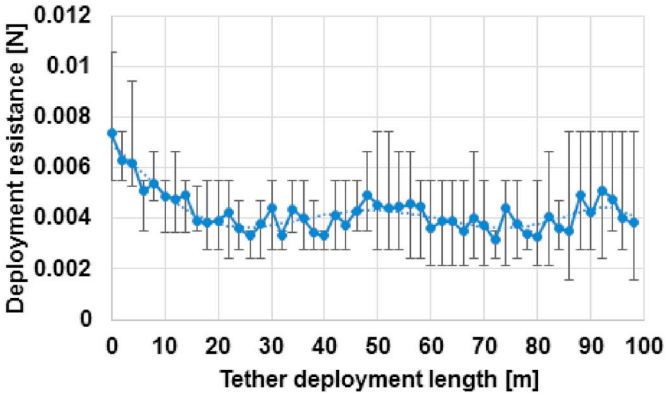


Fig. 17. Deployment resistance force of tether.

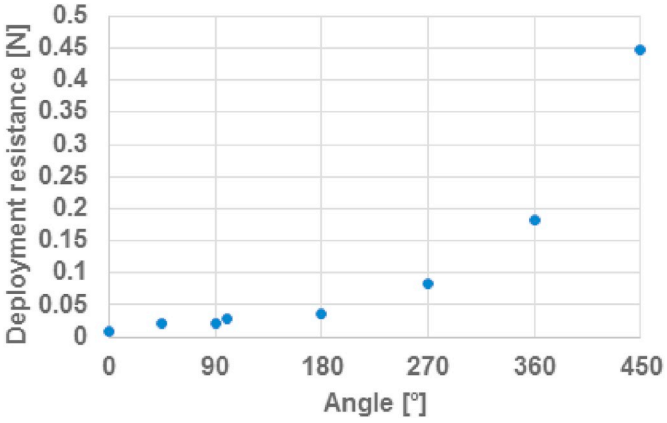


Fig. 18. Deployment resistance force of tether when satellite rotates.

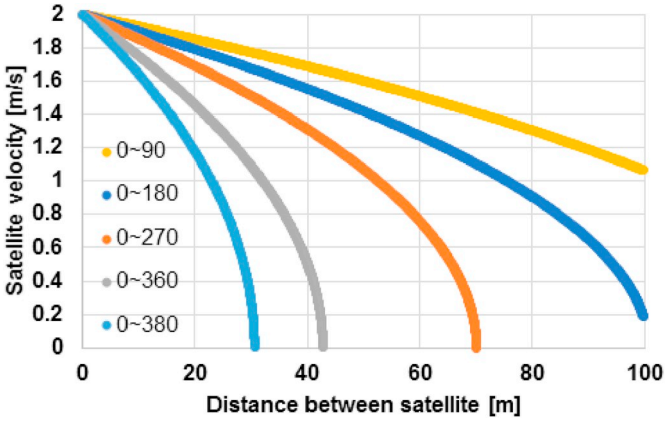


Fig. 19. Variation of satellite separation velocity with satellite distance when MS and DS rotate.

sequence of deployment. Once the MS and DS had rebounded, the tether loosened and MS and DS themselves again rotated because of the conservation of angular momentum, and the tether then wound around MS and DS. As the winding of the tether around the MS and DS advanced and the residual tether length became shorter than the distance between the MS and DS, the tether tension increased, the whole system of tethered MS and DS by tether rotated, and centrifugal force was again produced. This centrifugal force suppressed the rotation of the MS and DS and was balanced with the rotational force at a certain tether length. However, it is difficult to estimate strictly where the centrifugal force balanced with the rotational force from the obtained data.

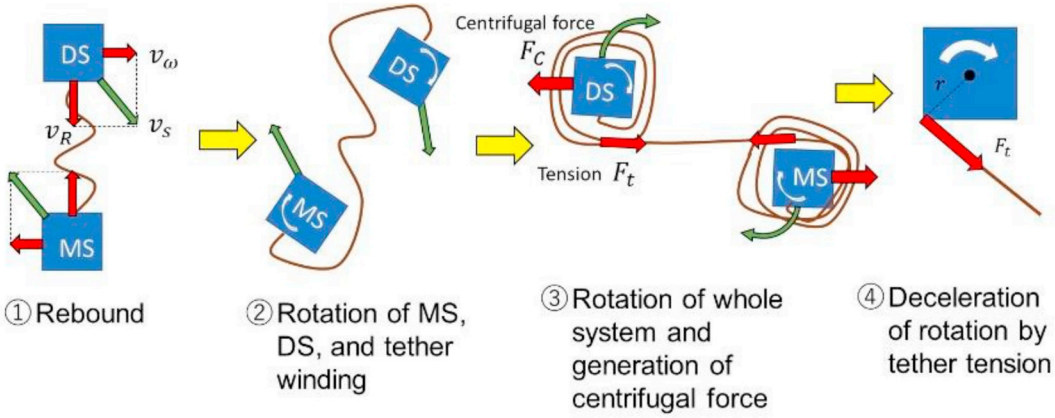


Fig. 20. Possible motion of tether and satellites after the tether was fully deployed and then wound around MS and DS.

7. Summary

STARS-C is a 2U-sized CubeSat developed to obtain data on tether deployment in orbit for the important technology of space elevators. STARS-C was launched on December 9, 2016, released from J-SSOD on December 19, 2016, and was operated until its re-entry on March 2, 2018. Unfortunately, detailed data on the tether deployment could not be obtained because the transmission conditions were not good. However, we found that STARS-C was rotated based on the optical data of STARS-C detected from the ground as supported by JAPOS, and this rotation significantly influenced the tether dynamics during and after deployment. We also estimated the tether deployment state from the data on the variation of STARS-C altitude, the satellite condition obtained by the optical data, and additional ground tests. As a result, it could be inferred that the tether was deployed about 30 m during the mission. There were two possible reasons why deployment was so limited. One was that tether deployment was stopped at about 30 m by the rapid decrease in deployment speed in order to increase the deployment resistance with the rotation of the MS and DS. The other was that the tether first deployed fully but then rebounded and then wound around the MS and DS by their rotation, and finally the centrifugal force produced by the rotation of the whole system connected MS and DS by the tether balanced with the rotational force of the MS and DS at a certain tether length. However, it is difficult to verify this latter scenario from the obtained data. In any case, it is surely considered that the rotation of the satellite had a large effect on the tether deployment condition in STARS-C.

Acknowledgement

We are grateful to JAXA for the opportunity to release STARS-C from Kibo. This opportunity enabled us to respond flexibly to the needs of users and to release a small satellite that could be realized only by Kibo which has unique equipment such as an air-lock mechanism and robot arm mechanism.

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Appendix A. Derivation of tether tension by centrifugal force with satellite rotation

If the masses of the MS and DS are *mm* and *md*, respectively, the distance of the MS and DS from the center of rotation is *rm* and *rd*, and the angular velocity of the tether connected system is ω, then the tension of the tether by centrifugal force acting on MS and DS, *FC*, is,

$$F_C = m_m r_m \omega^2 + m_d r_d \omega^2 \tag{A-1}$$

where $rm + rd = ltet$ (*ltet*: the deployed tether length) (Fig. A-1).

The angular velocity ω is given by the angular momentum L of rotational object as follows.

$$\omega = \frac{L}{m_m {r_m}^2 + m_d {r_d}^2} \tag{A-2}$$

From A-1 and A-2, the tension of the tether by centrifugal force, *FC*, can be rewritten as

$$F_C = [m_m r_m + m_d r_d]\left[\frac{L}{m_m {r_m}^2 + m_d {r_d}^2}\right]^2 \tag{A-3}$$

The analytical result shown in Fig. 15 was obtained by the simple assumptions that the mass of the MS and DS were the same, and that the MS and DS did not rotate themselves, namely, the angular momentum of STARS-C was transformed only to the rotation of the whole system in which MS and DS were connected by the tether. The conserved angular momentum L is calculated as follows by using mass, size and rotation data of STARS-C before the satellites separated.

$$L = \frac{m_m + m_d}{12}[l_w^2 + l_h^2]\omega_0 = \frac{2.5}{12}[0.\ 1^2 + 0.227^2]1.57 = 0.02\ (\mathrm{kgm^2/s}) \tag{A-4}$$

Appendix B. Measurement of tether deployment resistance

Figure B-1 shows the setup of the experiment to measure tether deployment resistance. First we marked the tether every 2 m and wound the whole 100 m tether around the spool, the same as we did with STARS-C. We then pulled out the spool from the wound tether and put the tether in a container with the same exit shape as used in STARS-C. The tether deployment resistance was determined as the value of the weight that the tether began to deploy when the mass of the weight was gradually increased. The resistance was measured every time 2 m of tether was pulled out.

To measure tether deployment resistance when the satellite rotated, we prepared an aluminum plate that imitated the surface of the satellite and simulated a situation in which the tether contacted the satellite surface. The data were obtained by changing the rotation angle of the satellite (rotation angle of satellite to the deployment direction of tether around the center of the satellite, θ in Fig. B-2) between 0 and 450°.

Appendix C. Measurement of spring force by tether coiling

As in the case of the measurement of tether deployment resistance, the wound-up 100 m tether marked every 2 m was put into the container, then the spring force of tether coiling was measured every 2 m as follows.

The spring force of tether coiling, F_S, is obtained by Eq. C-1,

$$F_S = ky \tag{C-1}$$

The spring constant by tether coiling k was obtained to measure the elongation of coiled tether y_m when the weight of mass m_w was put on the tether, and calculated from Eq. C-2 shown below. The measurement was performed by pulling out the tether every 2 m and fixing the tether at the exit of the container not to deploy by the weight (Fig. C-1).

$$k = \frac{m_w g_0}{y_m} \tag{C-2}$$

The total spring constant of deployed tether was determined as the spring constant of the series spring of every 2 m length for which spring the constant was measured (Fig. C-2).

The elongation of the coil shaped tether was determined as the difference between the natural length of the tether in the coiling condition and the tether length in the straight condition by measuring the natural length of tether l in the coiling condition (see Fig. C-1). The natural length of the tether in the coiling condition was measured for every 2 m of tether length in a straight condition (length between every marked point), then the spring force by tether coiling for a certain deployed length of tether was determined by Eq. C-1 using the total spring constant value and the sum of the elongation of every 2 m length of tether until the deployed length was reached.

Appendix D. Calculation method of tether deployment

The tether deployment condition was calculated by solving the equation of satellite motion. Figure D-1 shows the conditions of tether deployment. The equation of motion of the DS when MS was fixed is shown as Eq. (D-1). The velocity of the DS at distance x is shown as Eq. (D-2).

$$m_d \frac{d^2x}{dt^2} = F_g + F_C - F_R - F_S \tag{D-1}$$

$$v_d = \frac{dx}{dt} \tag{D-2}$$

F_C is the tether tension by centrifugal force with satellite rotation, F_R is the tether deployment resistance, and F_S is the spring force by tether coiling shown in Appendix A, B, and C and are given as data for tether deployment length. F_G is the gravity gradient force, and is calculated by the following equation when the tether is deployed in the Earth radial direction [16],

$$F_G \cong 3m_d l_{tet}\omega_s{}^2 \tag{D-3}$$

In the calculation in Fig. 19, we considered only the tether deployment resistance because the deployment resistance was an order of magnitude greater than other forces. We gave a 2 m/s initial velocity of the DS which was obtained by the springs housed in the satellite for the satellite separation and the experimental value by the ground test. We then calculated the relationship between the tether deployment length and the velocity of DS by the Fourth-Order Runge-Kutta method.

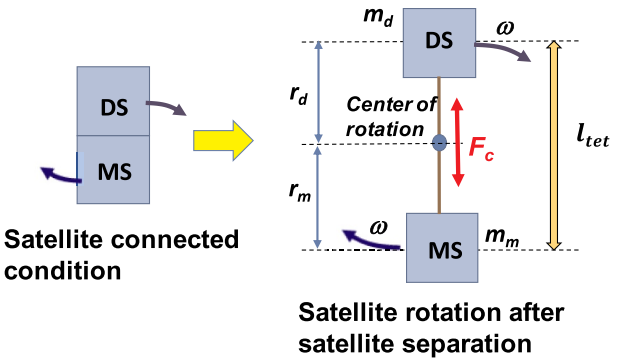


Fig. A1. Conceptual figure of tether tension by total satellite system rotation.

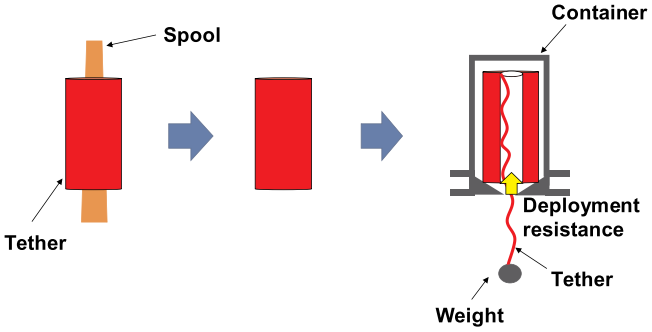


Fig. B1. Setup of experiment for measuring tether deployment resistance.

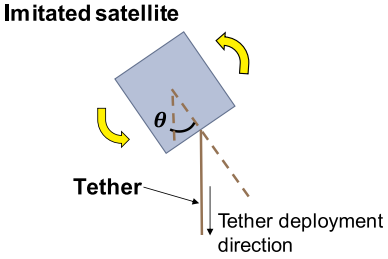


Fig. B2. Measurement of tether deployment resistance when the satellite rotates.

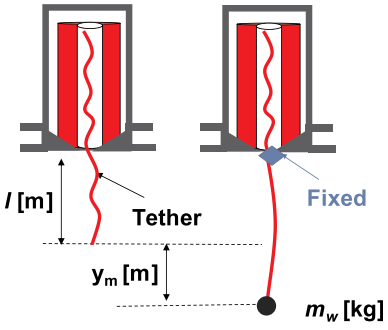


Fig. C1. Setup of experiment for measuring the spring force produced by tether coil shape.

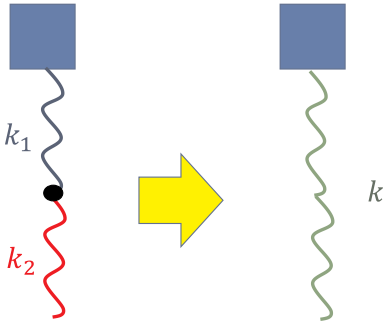


Fig. C2. Method to derive total spring constant of tether pulled out from each data of spring constant every constant tether length.

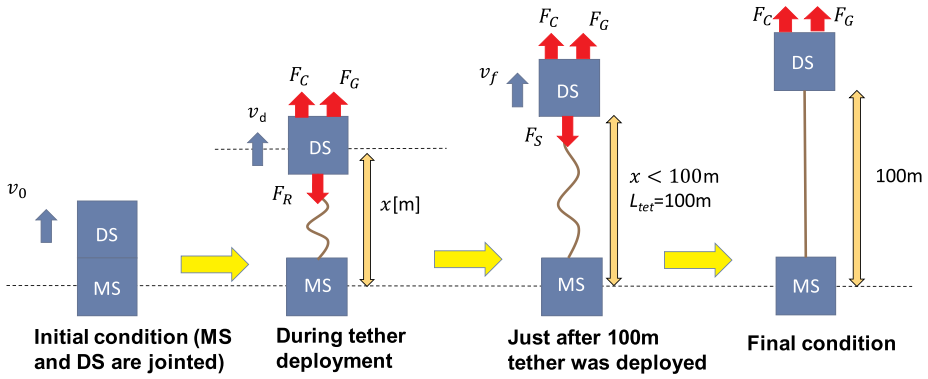


Fig. D1. Tether deployment condition.

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